LINEAR REGRESSION REPORT

I. Dataset

The Hungarian Chickenpox Cases dataset obtained from UCI Machine Learning Repository was used for the regression task [1]. It includes 521 cases under 20 features consisting of different regions in Hungary, such as Budapest, Baranya, Bacs, Bekes, and others.

II. Regression Result

Using the Scikit-learn library, linear regression was applied on the relationship between Baranya and other regions (name_2).

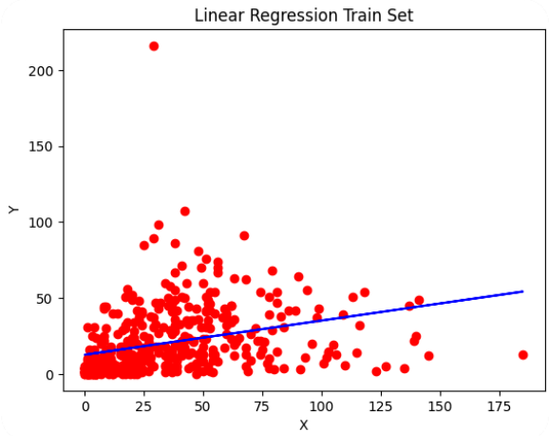


Figure 1. Train Test Set

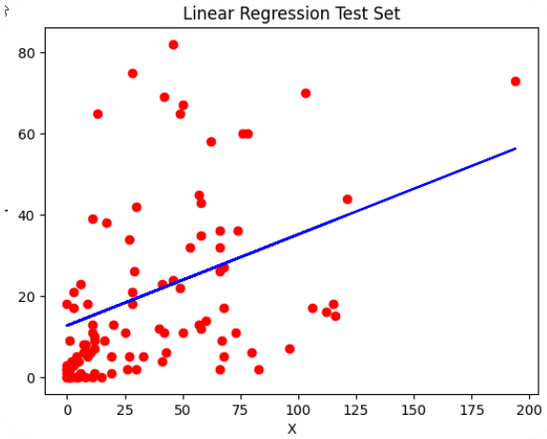


Figure 2. Regression Test Set

Figures 1 and 2 illustrate a positive correlation as revealed by the upward slope of the regression line, indicating a direct relationship between the regions. This suggests that frequent movement of people between these regions, potentially facilitating disease spread.

```
Mean Squared Error: 340.57889216273173
```

Figure 3. Mean Squared Error Value

The Mean Squared Error value MSE) for the test set is 340.5790. MSE measures the average squared difference between predicted and actual values. This indicates moderate to high prediction inaccuracy.

The high MSE suggests the linear regression model is not adequately capturing the underlying patterns in the data. Improving feature engineering or adopting more flexible models would likely enhance performance.

III. References

[1] Unknown. 2021. Hungarian Chickenpox Cases. https://doi.org/10.24432/C5103B